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MILITARY SPECIFICATION

ROCKET MOTOR, 2.75 INCH ROCKET, MARK 40 AND MODS, MARK 4 AND MODS

LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings, for one type of rocket motor designated as 2.75 Inch Rocket Motor Mark 40 Mod 3 and Mark 4 Mod 10.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition and Special Weapons General Specification For
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1340

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STANDARDS

FEDERAL

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

* 2.1.2 Other Government documents, drawings, publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

* ARMY

* 8883478	Cont. Ammunition Fiber, M518, For Rocket Motor, 2.75 Inch Mark 40 and Mods and Mark 4 and Mods
* 8883479	Box, Packing Ammunition For Rocket Motor, 2.75 Mark 40 and Mods and Mark 4 and Mods.
* 9220803	2.75 Inch Rocket Motor, Mark 40 and Mods Loaded Assembly
* 9220804	2.75 Inch Rocket Motor, Mark 4 and Mods, Loaded Assembly

PUBLICATIONS

ARMY

ET-9220803	Equipment Tabulation
ET-9220804	Equipment Tabulation

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* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.3 Order of precedence. In the event a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Loaded Assemblies. The loaded assemblies shall comply with all of the requirements specified on drawing (dwg) 9220803 or (dwg) 9220804 and with all requirements specified in applicable specification.

3.3 Electrical Continuity and Resistance. Each loaded motor ignition circuit shall be continuous and conform to the resistance values of 0.7 to 2.0 ohms.

* 3.4 Fluoroscope Inspection. Each loaded motor shall be fluoroscopically examined to verify the presence and proper condition, location and orientation of each component. The forward exposed live lead wire as it exits the piston of the nozzle and fin assembly shall be observed to assure that this wire does not exceed a bend or deformation of more than thirty degrees from the horizontal. The rocket motor shall be free from foreign material. "O" rings and seal rings shall show no evidence of cuts, twist or other damage resulting from assembly.

3.5 Ballistic Test. The loaded motor shall be field fired from a launcher with a detent release force of 200 ± 20 pounds. The launcher shall be approved by the procuring activity. Wind velocity and direction at launcher at time of firing shall be recorded for each round.

3.5.1 External Ballistics. The trajectory of each field fired motor shall be within a lateral deflection of plus or minus 150 mils at each of the firing temperatures. The lateral deflection is measured by the puff of black smoke caused by propellant grain burn out. If the burnout is not detectable the point of impact may be used or any other method approved by the procuring activity.

The range limits are as follows:

Temperatures, °F.	Range (Yards)	
	Lower	Upper
- 65	1500	7800
+115	2000	7800
+150	2500	7800

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3.5.2 Time of first motion. Delay of time of first motion at each temperature shall be as follows:

Temperatures, °F.	First Motion Time (Sec) Upper Limit
-65	0.120
+115	0.095
+150	0.085

3.5.3 Internal ballistics. Any loaded motor firing that results in a motor tube rupture or loss of components will cause the lot represented to be rejected.

3.5.4 Erratic behavior. The term erratic behavior is used to describe incidents that occur during field firings which in the judgment of the procuring activity might cause the lot to be suspect from a safety or performance standpoint. These include abnormal noise during propellant burning, a corkscrewing trajectory (see 6.5), tumbling (see 6.6) and other deficiencies which indicate the motor is not performing satisfactorily.

3.6. Static Firing. The loaded motor shall be static fired from a test stand. The test stand and recording equipment shall be approved by the procuring activity.

3.6.1 Early Maximum Thrust at 165°F. All loaded motors fired at +165 degrees fahrenheit (F) shall not exceed 1475 pounds thrust. Early maximum thrust is defined as the highest maximum thrust reached during the first 0.20 seconds of burning. If the trace oscillates, it is the highest thrust reached by the line connecting the midpoints of the initial thrust oscillations.

3.6.2 Late Maximum Thrust at +165°F. All loaded motors fired at +165 degrees fahrenheit (F) shall not exceed 1210 pounds thrust. Late maximum thrust is defined as the highest thrust reached after 85 percent of the burning time.

3.6.3 Incipient Failure. All motor tubes after static firing shall be free of any incipient failures that result in a bulge in the side of the motor tube 1/16 inch high or higher.

3.6.4 Thrust Time Curves. Thrust time curve traces shall lie entirely between the appropriate dotted lines of Fig. 1.

3.6.5 Ignition Delay. The maximum allowable ignition delay which is defined as the time from zero (0) time (current applied to igniter) to the 150 pound thrust level, shall not exceed the values shown below:

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Firing Temperature, °F	Maximum Ignition Delay Time (Milliseconds)
-65	110
+165	50

3.6.6 Internal ballistic (static). Any loaded motor firing that results in a motor tube rupture will cause the lot represented to be rejected.

3.7 Resistance of shorted motor. Each motor, after the shorting clip has been installed, shall have a resistance of no more than .09 ohms between the Contact Disc and the Nozzle Plate.

3.8 Leak test of the loaded motor. The loaded motor in the final assembled configuration shall be subjected to a leak test. Each loaded motor shall be capable of withstanding an applied external pressure of ten (10) PSI without exceeding a leakage rate of ten (10) to the minus forth cubic centimeter per second. (10-4 c.c/sec.).

3.9 Workmanship. All components shall be fabricated and assembled in accordance with best practices. They shall be free of all foreign matter. Metal parts shall be free of rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. Surface coatings shall be continuous except for a few light scratches which do not expose base material.

3.10 First Article Inspection. This specification makes provisions for first article inspection. Requirement for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of Product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspection.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind has principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification. Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

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4.2.1 Submission. The contractor shall submit a sample of 100 assemblies to the Government for tests in accordance with the provisions of 4.2.2. All parts and material shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirement of the contract.

4.2.2 Inspections to be Performed. Assemblies and components will be submitted by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, except that para. 4.3.3.3 will be sixty (60) in lieu of twenty-four (24) loaded motor assemblies, and para. 4.3.3.4 will be forty (40) in lieu of sixteen (16) loaded motor assemblies, and any or all requirements of the applicable drawings.

4.2.3 Rejection. If any assembly or component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions for production

4.3.1 Lot formation. The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantitites presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to either the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering as required, shall be in accordance with MIL-STD-1168.

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The maximum number of lots of components and the source of each component lot which can be assembled into one loaded motor lot is as follows:

Component	Maximum Number of lots	Source of component lots
Motor Tube	Three	A single Contractor's production
Nozzle & Fin Assembly	Six	A single Contractor's production
Igniter	Two	A single Contractor's production
Propellant grain	One	A single Contractor's production
Seal Ring	Two	A single Contractor's production
"O" Ring	Two	A single Contractor's production
Metal Spacer	Two	A single Contractor's production
Stabilizing Rod	Two	A single Contractor's production
Charge Support Disc	Two	A single Contractor's production
* Charge Support Spacer	Four	A single Contractor's production
Charge Support Ring	Two	A single Contractor's production
Wire Lock	Two	A single Contractor's production
Felt Washer	Two	A single Contractor's production

a. There will be no intermingling of components between manufacturers except as noted above. Insofar as practicable, manufacturers lot numbers within an ammunition lot and in consecutive ammunition lots will be in consecutive numerical sequence.

b. Every effort will be made to assemble components of metal parts lots together.

c. Lot numbers and quantities for each of the listed metal parts components shall be recorded on the reverse side of the data card.

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Unless otherwise approved by the Contracting Officer, the inspection lot size of motors deliverable under contract shall be approximately 20,000 units. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogenous and of a size convenient to the contractor and the Government Inspector.

4.3.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.
* AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's (classification of defects) listed in this specification. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 Rocket Motor, 2.75 Inch, Mark 40 Mod 3 loaded assembly (see dwg 9220803), or Rocket Motor, 2.75 Inch, Mark 4 Mod 10 loaded assembly (see dwg 9220804).

Cateorgies	Defects	Methods of Inspection	Code No. (See 6.2)
Critical:			
1.	Shorting clip missile or making improper contact	Visual/ Manual	01001
Major:	AQL 1.0 percent		
101.	Live and ground lead wires assembled to igniter wires improperly	Visual/ Manual	01002
102.	Number of spacers incorrect	Visual	01003
103.	Lockwire installed in incorrect direction	Visual	01004
* 105.	Lockwire end not flush or below O.D of tube	Visual/ Manual	01006
106.	Painting or marking missing, incorrect, or illegible	Visual	01007

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Minor:	AQL 0.65 percent		
201.	Component lubrication missing or improper	Visual	01008
202.	Evidence of poor workmanship (see 3.9)	Visual	01009

* 4.3.2.2 Cont. Ammunition, Fiber, M518 for Rocket Motor; 2.75 Inch, Mark 40 and Mods and Mark 4 and Mods (See Dwg. 8883478).

Categories	Defects	Medthod of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.065 Percent		
101.	Contents exposed	Visual	02001
102.	Contents loose	Visual/ Manual	02002
103.	Fillers missing or inadequate	Visual	02003
104.	Fin Separater Assembly improperly positioned or inadequate	Visual/ Manual	02004
Minor:	AQL 0.65 Percent		
201.	Marking, misleading or unidentifiable	Visual	02005

* 4.3.2.3 Box, Packing, Ammunition for Rocket Motor; 2.75 Inch Mark 40 and Mods and Mark 4 and Mods, prior to sealing (see dwg. 8883479).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Descriptive nomenclature incorrect	Visual	03001
102.	Contents positioned improperly	Visual	03002
103.	End, side and top fillers missing or inadequate	Visual	03003
Minor:	AQL 0.065 Percent		
201.	Foreign material in box	Visual	03004

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* 4.3.2.4 Box, Packing, Ammunition for Rocket Motor: 2.75 Inch Mark 40 and Mods and Mark 4 and Mods (see dwg 8883479).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.0 percent		
101.	Contents exposed	Visual	04001
102.	Board broken	Visual	04002
103.	FSC number missing or misleading	Visual	04003
104.	Hardware missing, broken or loose	Visual/ Manual	04004
105.	Strapping missing, broken or loose	Visual/ Manual	04005
Minor:	AQL 1.50 percent		
201.	Marking, misleading or unidentifiable	Visual	04006
202.	Hardware improperly engaged	Visual	04007
203.	Car seal missing, unsealed or improperly positioned	Visual	04008
204.	Handle missing or insecure	Visual/ Manual	04009

4.3.3 Testing.

* 4.3.3.1 Continuity test (see 3.3) Major Defect Code No. 01010. A sample in accordance with MIL-STD-105 shall be selected using an AQL of 0.015 percent and Inspection Level II. These samples shall be subjected to the test as specified in 4.4.1 using equipment in accordance with 4.3.4. Any motor not meeting the resistance requirement shall be considered a reject.

4.3.3.2 Fluoroscopic Test (see 3.4) Major Defect - Code No. 01011. To insure that the loaded motor assembly is in the proper configuration as designed, each lot of loaded motor assemblies shall be one hundred percent subjected to a fluoroscopic test as specified in 4.4.2 using equipment in accordance with 4.3.4 Any motor that does not meet the requirements of 3.4 shall be removed from the lot.

4.3.3.3 Ballistic Test (see 3.5) Internal Ballistics (see 3.5.3) Critical Defect - Code No. 01012; External Ballistics, Time of First Motion, and Erratic Behavior (see 3.5.1, 3.5.2, 3.5.4) Major Defect Code No. 01013, 01014, 01015, respectively.

To insure that the loaded rocket motor assembly will function as designed, a ballistic test sample of twenty-four (24) loaded motor assemblies shall be randomly selected from each production lot.

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The samples shall be equally divided, fired at the three temperatures specified in 3.5.1 and shall meet the requirements specified in 3.5, when tested as specified in 4.4.3 using equipment in accordance with 4.3.4. A visual examination of each round shall be performed and recorded to determine any anomalies that may exist on the external surfaces of the motor body and the nozzle and fin assembly prior to firing. Any * anomaly shall be recorded.

An inert loaded M151 Warhead with an insert M423 Fuze or Head and Fuze combination approved by the Procuring Activity shall be temperature conditioned with the loaded motor assembly. Each loaded motor ignition circuit shall be continuous and recorded prior to firing. If any round fails the internal ballistic requirements, the lot shall be rejected.

If two rounds or more fail to pass either the requirements for external ballistics, or time of first motion, or erratic behavior the lot shall be rejected. If one round fails to pass either external ballistics or time of first motion or erratic behavior, the lot will be retested with a double sample at each temperature. Any one failure to the requirements of internal ballistics, external ballistics, erratic behavior, or time of first motion during retesting shall reject the lot.

Firing of the entire sample shall be completed regardless of when the rejection number occurs. The remaining assemblies or parts of any round that is rejected shall be * gathered and kept for evaluation. A record of test results shall be sent to the address specified in the contract. If temperature recycling occurrs, this will be noted on the record of test.

4.3.3.4 Static Fire Test (see 3.6). Internal Ballistics, Incipient Failure (see 3.6.6, 3.6.3) Critical Defect - Code No.01016, Code No. 01017 respectively; - Early Maximum Thrust, Late Maximum Thrust, Thrust Time Curves, Ignition Delay (see 3.6.1, 3.6.2, 3.6.4, 3.6.5) Major Defect - Code No. 01018, 01019, 01020, 01021 respectively.

To insure that the loaded rocket motor assembly will function as designed, a static fire sample of sixteen (16) loaded motor assemblies randomly selected from each of the first five (5) production lots; and every fifth lot thereafter shall be static fired as specified in 4.4.4 using equipment in accordance with 4.3.4. In the event of failure of a lot, samples from each succeeding lot shall be static fired until a five successive lots have successfully passed all tests, then each fifth lot thereafter shall be static fired. Each static firing sample shall be visually inspected prior to firing and any anomalies shall be recorded. Each static firing sample will be x-rayed prior to firing in two planes and shall be identifiable to the appropriate motor for any future reference.

The sample shall be equally divided; conditioned at the two temperatures, -65°F. and +165°F. and static fired. Representative lot is to be rejected if early maximum thrust exceeds the limits as specified in 3.6.1 on two (2) motor firings. Representative lot is to be rejected if late maximum thrust exceeds the limits as specified in 3.6.2

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on two (2) motor firings. Representative Lot is to be rejected if ignition delay exceeds the limits as specified in 3.6.5 on two (2) motor firings. Representative lot is to be rejected if incipient failure as specified in 3.6.3 occurs on one (1) loaded motor firing. Representative lot is to be rejected if a motor tube repture as specified in 3.6.6 occurs on one (1) motor firing. Representative lot is to be rejected if thrust time curves (Fig. 1) do not conform to the limits on the curves of Fig. 1 on three (3) loaded motor firings.

If one (1) loaded motor firing fails for early maximum thrust, or late maximum thrust, or ignition delay, or 1 or 2 thrust time curves do not conform to Figure 1, the lot shall be retested with a double sample at each temperature. Any one failure during retesting to early maximum thrust, or late maximum thrust, or incipient failure, or ignition delay, or the thrust time curves, or internal ballistics shall fail the lot. Firing of entire sample is to be completed regardless of when the rejection number occurs. A record of test results shall be sent to the address specified in the contract. * Any rounds indicating blisters or burn marks shall be noted on the firing record. Any round reconditioned shall be noted on the firing record.

* 4.3.3.5 Resistance of shorted motor (see 3.7. Critical Defect - Code No. 01022. Each motor with the shorting clip in place from each lot will be subjected to a resistance check as specified in 4.4.5 using equipment in accordance with 4.3.4. One failure or more will reject the lot.

*

4.3.4 Inspection equipment. Equipment Tabulation Number ET 9220803 or ET 9220804 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DoD activities for design, procurement, manufacture, testing, evaluation, production, and receiving inspection, overhaul, shipping storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance. Supply and maintenance of the equipment listed on the ET shall be in accordance with MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Continuity test (see 4.3.3.1). Check continuity of ignition circuit between contact disc and nozzle plate. Electrical resistance of this circuit shall not be less than 0.7 ohms or more than 2.0 ohms with the shorting clip removed. The test current used shall be 25 milliamperes or less (.025 amps).

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4.4.2 Fluoroscopic test (see 4.3.3.2). The loaded motors shall be slowly rotated about their longitudinal axis during viewing. The technique shall be judged correct when the image details of the component parts are sharply defined on the fluoroscopic screen.

4.4.3 Ballistic test (see 4.3.3.3). Each loaded motor selected shall be conditioned to assure a uniform temperature throughout its entire mass at the time of firing. Minimum conditioning time shall be eight hours; maximum conditioning time shall be 24 hours. During the last three hours of conditioning, the cabinet temperature shall be maintained within three degrees of the temperature specified in 3.5.1. Each firing shall occur within ten minutes from the time the loaded motor is removed from the temperature-conditioning cabinet. A motor left out of the conditioning cabinet over ten minutes prior to firing shall be placed at ambient temperature for not less than eight hours and then reconditioned to the applicable temperature. Recording equipment used for measuring lateral deflection and range shall be approved by the procuring activity. A direct current (dc) power supply shall provide a firing pulse of 2.8 to 3.2 amperes to the ignition circuit of the loaded motor. Firing of the entire sample shall be completed regardless of when the rejection number occurs. Firing shall be conducted at a launch quadrent elevation of 12.5 to 13.5 degrees. True Azimuth and velocity of the wind shall be recorded for each round fired.

4.4.4 Static test (see 4.3.3.4). Each loaded motor selected shall be conditioned to assure a uniform temperature throughout its entire mass at the time of firing. Minimum conditioning time shall be eight hours. Maximum conditioning time shall be 24 hours. During the last three hours of conditioning the cabinet temperature shall be maintained within three degrees of the specified temperature. Any loaded motor not fired within ten minutes shall be returned to the applicable temperature conditioning cabinet for not less than one hour. A motor left out of the oven over ten minutes prior to the firing shall be placed at ambient temperature for not less than 12 hours before reconditioning to the applicable temperature.

The motor thrust adapter, thrust load cell, and recording equipment used to measure the performance data shall be approved by the procuring activity. A dc power supply shall provide a firing pulse of 2.8 to 3.2 ampers to the ignition circuit of the loaded motor. The resistance of the loaded motor shall be determined and recorded prior to firing.

Firing of the entire sample shall be completed regardless of when rejection number occurs. Thrust time curve traces shall lie entirely between the appropriate dotted lines of Figure 1. All thrust time curves shall be examined. Each curve shall be classified as being normal, or defective. The condition which would cause a thrust time curve to be classified as defective is failure of the curve to meet the shape requirements of Figure 1. A sudden drop in thrust on the time curve which exceeds 150 pounds followed by an abrupt return to a normal thrust level calls for a thorough investigation as to the cause.

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4.4.5 Resistance of shorted motor (see 4.3.3.5). Each sample motor shall be subjected to a resistance check with the shorting clip in place. The motor probes shall be placed one on the contact disc (the probe is not to contact the shorting clip) and one on the nozzle place outside diameter (the probe is not to contact the shorting clip) and the resistance read. The resistance shall not be more than .09 ohms. The equipment used shall not subject the circuit to more than 25 milliamperes (.025 amps.).

4.4.6 Leak test (see 4.3.3.6). The motors selected for test shall be subjected to a leak test using nitrogen gas as the working fluid. The loaded motor when subjected to a pressure of ten (10) PSIG at room temperature (68°F.) shall not leak at a rate to exceed ten to the minus forth cubic centimeters per second (10-4 cc/second). Under test, the motor shall remain within limits for ten (10) seconds minimum. The leakage rate shall be recorded.

* 5. PACKAGING

5.1 Packaging. Level A

5.1.1 Motors. Packaging shall be in accordance with drawing 8883478.

5.2 Packing. Level A

5.2.2 Motors. Packing shall be in accordance with drawing 8883479.

5.3 Marking. Unit and exterior containers shall be marked in accordance with marking instructions contained in the applicable drawings.

* 5.3.1 Motors. Marking requirements for the MK40 Mod 3 are as follows:

2.75 INCH ROCKET MOTOR MOD 3
(LOT NO) YEAR

* 5.3.2 Motors. Marking requirements for the MK 4 Mod 10 are as follows:

2.75 INCH ROCKET MOTOR MK4 MOD 10
(LOT NO) (YEAR)

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

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6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army design. Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, * Level 2. Legibility and reproducibility shall permit conventional making of clearly

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understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

* 6.3.5 Submission of contractor designs. Submission of contractor designs for approval. All submitted designs shall identify the defect(s) the equipment has been designed to check by referencing the applicable five digit code number, contained in Section 4 of this specification, and the appropriate component or assembly drawing number to which the specific design applies. Unless otherwise specified on the EL, all designs and subsequent design changes of equipment for inspection of defects classified as critical and major shall be submitted for approval to the address specified in the contract. Design review will be completed normally within one month after receipt of the design. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, completion date for design review will be based on the date of the final submission of designs.

* 6.4 Submission of First Article Samples. Instruction as to the location for the evaluation of the first article sample shall be obtained from the Contracting Officer. All inquiries should be forwarded to the address specified in the contract.

6.5 Corkscrewing. Corkscrewing shall be defined as any spiral movement or displacement of the aft end of the motor about the predicted straight-line trajectory of the rocket during flight. A round to be considered defective for corkscrewing shall exhibit an angle of 30^{0} or more between the normal flight path and the furthest displacement of the aft end of the motor. Observance of the round and not of the smoke will be used to determine corkscrewing. This displacement shall be observed and recorded by a trained and experienced range observer. Care should be exercised when scarfed nozzle motors are fired because these motors are designed to rotate during flight and may appear to an untrained observer to corkscrew. Range wind conditions should also be noted and observed because this factor may also lead to observations that may appear to be corkscrewing smoke trails.

6.6 Tumbling. Tumbling shall be defined as any end - over - end displacement of the motor during flight.

* 6.7 Report of testing. The results of the Ballistic Test (Table 1), Static Fire Test (Table 2), Leak Test (Table 3), shall be submitted to the address specified in the contract.

6.8 Failure analysis. A brief statement of any failure occuring during testing (Table 1, Table 2, Table 3) shall be submitted by the Contractor. This statement should reflect any observation or fact that will be helpful to the procuring activity when a more comprehensive analysis, based partically on this failure analysis, is made. The Government representative shall indicate his concurrence or comments on the analysis.

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* 6.9 Changes from previous issue. The margins of this specification are marked with codes to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-MI

Preparing Activity:
Army-MI

Project No. 1340-A470

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TABLE I

BALLISTIC TEST (SEE 3.5)

CONTRACTOR: _____ REFERRED: _____

MOTOR LOT NO. _____ REFERRED FOR RETEST: _____

QUANTITY: _____ REJECTED: _____

DATE: _____ ACCEPTED ON RETEST: _____

TYPE HEAD: _____ REJECTED ON RETEST: _____

TYPE FUZE: _____ OTHER: _____

ACCEPTED: _____ ACCEPTED WITH WAIVER: _____

RD. NO.	RD TEMP	WIND		TIME OF FLIGHT (SEC)	RANGE		DEFLECTION		RD. TEMP. RECYCLED	TIME TO BURNOUT (SEC)
		DIR. (DEG)	VEL (KTS)		OBSERVED (YDS)	EST. (YDS)	OBSERVED (YDS)	EST. (YDS)		

BALLISTIC TEST (SEE 3.5)

RD. NO.	RD. TEMP	TIME OF FIRST MOTION (SEC)	ERRATIC BEHAVIOR	INTERNAL BALLISTICS	X-RAY RESULTS	CIRCUIT RES. (OHMS)	VISUAL ANOMALIES	RANGE AT BURNOUT

Atmospheric Conditions: TEMP (Deg. F) _____ Pressure (Inches Hg) _____

BALLISTIC TEST (SEE 3.5)

Serial Number of Defect	Failure Analysis (See 6.8)

_____
Government Representative

_____
Contractor Representative

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TABLE 2

CONTRACTOR: _____

MOTOR LOT NO: _____

QUANTITY: _____

DATE: _____

REFERRED: _____

REFERRED FOR RETEST: _____

REJECTED: _____

ACCEPTED ON RETEST: _____

REJECTED ON RETEST: _____

OTHER: _____

ACCEPTED: _____

ACCEPTED WITH WAIVER: _____

Static Fire Test (see 3.6)

RD. NO.	RD. TEMP	EARLY MAX THRUST (LB.)	LATE MAX THRUST (LB.)	THRUST TIME CURVE	INCIPIENT FAILURE	IGNITION DELAY (SEC)	INTERNAL BALLISTIC	X-RAY RESULTS	VISUAL	
									BEFORE	AFTER

Static Fire Test (see 3.6)

Round Number	Failure Analysis (6.8)

_____
Government Representative

_____
Contractor Representative

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TABLE 3

CONTRACTOR: _____

MOTOR LOT NO: _____

QUANTITY: _____

DATE: _____

REFERRED: _____

REFERRED FOR RETEST: _____

REJECTED: _____

ACCEPTED ON RETEST: _____

REJECTED ON RETEST: _____

OTHER: _____

ACCEPTED: _____

ACCEPTED WITH WAIVER: _____

LEAK TEST (SEE 3.8)

RD. NO.	LEAKAGE RATE	FAILURE ANALYSIS (SEE 6.8)

_____
Government representative

_____
Contractor representative

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TABLE 4

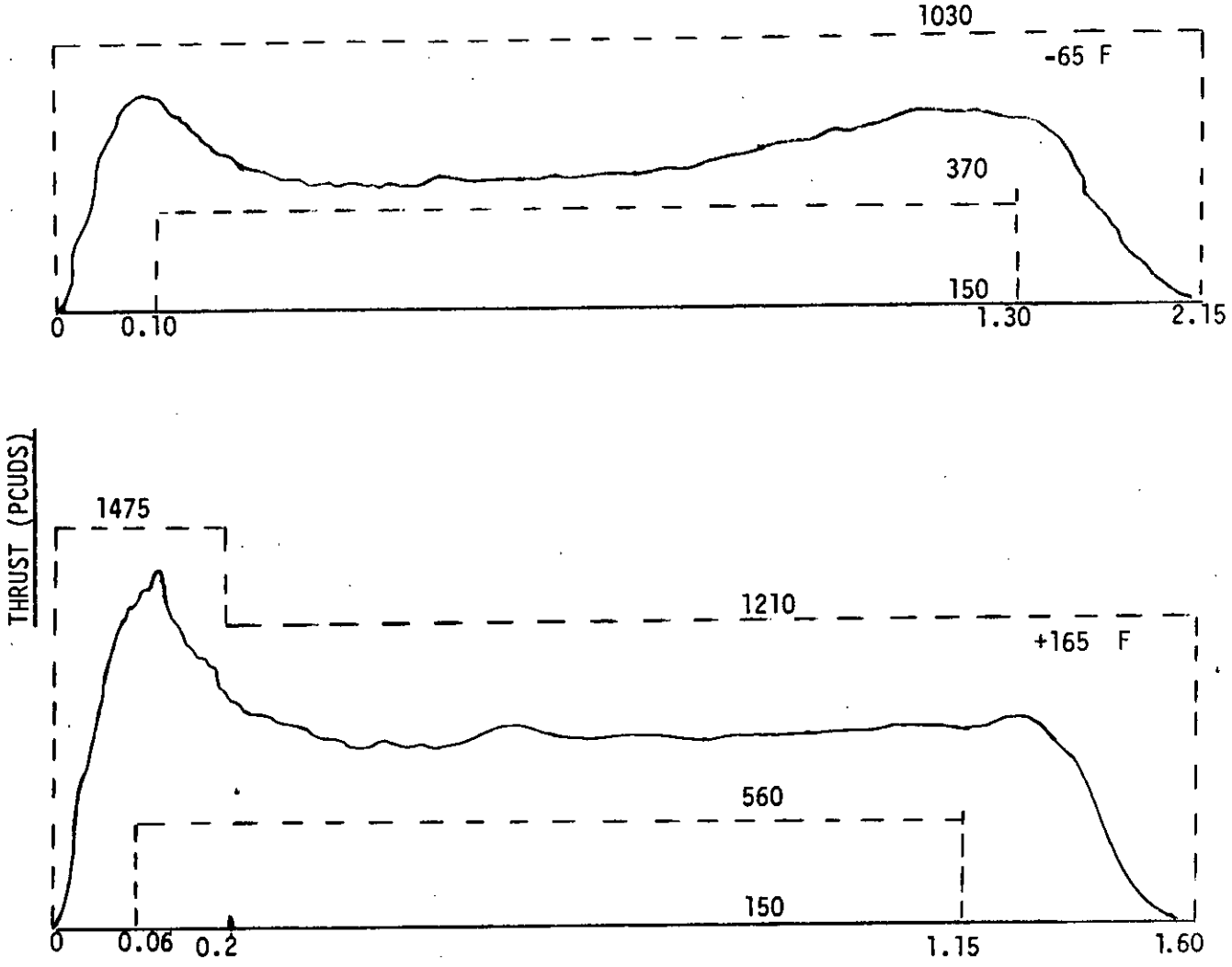
Acceptance test paragraph number		Number for rejection of lot	Number for retest
4.3.3.3	Field firing		
4.3.3.3	Internal ballistics	1	
4.3.3.3	Erratic behavior	2	1
4.3.3.3	External ballistics	2	1
4.3.3.3	Time of first motion	2	1
4.3.3.4	Static Firing		
4.3.3.4	Early maximum thrust at +165°F	2	1
4.3.3.4	Late maximum thrust at +165°F	2	1
4.3.3.4	Incipient failure	1	
4.3.3.4	Ignition delay	2	1
4.3.3.4	Thrust time curves	3	1 or 2
4.3.3.4	Internal ballistic (static)	1	

Summary of action to be taken for nonconformity of requirements of Paragraph 3.5.3.6 for number of failures indicated in each column shown above.

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FIGURE 1

THRUST TIME CURVE LIMITS FOR 2.75 INCH
ROCKET MOTOR
MARK 40 AND MODS, MARK 4 AND MODS



*TIME (SECONDS)

*This starts from application of firing pulse
Times and thrust are not to scale

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-R-60995A Rocket Motor, 2.75 Inch Rocket, Mark 40 and Mods, Mark 4 and Mods Loading, Assembling and Packing

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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